Darren Lee

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Education

University of California, Berkeley | *Regents' and Chancellor's Scholar* Expected Graduation: May 2029

B.S. Electrical Engineering and Computer Science, Minor in Materials Science and Engineering *Berkeley, CA*

- **Relevant Coursework:** Computer Architecture, Circuits and Devices, Properties of Materials, Signals and Systems, Hands-On PCB Engineering, Data Structures, Linear Algebra + Diff Eq, Physics for Scientists and Engineers, SICP

Skills

Hardware: Circuit design, Circuit simulation (LTspice), PCB design (KiCad), Digital logic design (Logisim), High-voltage design, RF design, Transformer design, Bench instrumentation, Embedded systems, CAD (SolidWorks), Prototyping, Arduino
Software: C, C++, Python, RISC-V assembly, Java, SQL

Experience

Evan R. Williams Research Group (via Engineering Solutions at Berkeley) Feb 2026 – Present

Electrical Engineering Consultant, RF Generator for Mass Spectrometry *Berkeley, CA*

- Developing a custom high-voltage RF generator for a charge-detection mass spectrometer, targeting 1 to 1.5 kVpp quadrupole and/or 250 Vpp ion funnel drives from 10 kHz to 1 MHz into 300 pF to 1 nF capacitive loads for precise ion confinement
- Currently designing the high-voltage resonator PCB, a series LC tank tuned by 7 relay-switched inductor sections per side (5 µH to 3.2 mH) and a 2 pF to 20 nF capacitor bank to reach 750 V amplitude per output on a 1 kV DC offset
- Custom-wound and swept toroid and U-core transformer prototypes from 50 kHz to 3 MHz, selecting a Material 43 NiZn toroid that delivered 474% higher step-up gain at 1 MHz than a Material 77 MnZn U-core under identical 1:4 winding
- Benchmarked 5+ commercial step-up transformers against the original 100 Hz to 10 MHz spec, finding none met turns ratio, voltage, and bandwidth together and driving the move to a custom-wound design

Fusion Dynamics June 2026 – Aug 2026

Electrical Engineering Intern *Oakland, CA*

- Scoped a drone-agnostic autonomous docking station that concentrates landing intelligence in the dock rather than the aircraft, evaluating fiducial-marker landing, IR beacon tracking, and Betaflight command uplink for pilot-free landing
- Designed a four-LED 850 nm IR beacon board in KiCad, computationally comparing emitter geometries and selecting a layout with nearly 3x the rotational distinctness of the initial candidate so the pose solver could not lock onto a wrong orientation
- Reviewed a 4-layer 95 x 55 mm structural frame PCB doubling as the power distribution board of a sub-250 g drone, catching a connector pin shortage and a 2 oz copper vs. controlled-impedance conflict before fabrication

Malin Space Science Systems (via Engineering Solutions at Berkeley) Sept 2025 – Dec 2025

Electrical and Software Engineering Consultant, Precision Focus Actuator *Berkeley, CA*

- Designed and prototyped a focus actuator for space camera applications used by NASA rovers, delivering 12.96 mil of travel (3.2x the ±2 mil range spec) at about 4.9 nm per step across 3 printed builds and a CNC-machined build
- Developed a bidirectional homing and midpoint-referencing algorithm using a DRV8834 microstepped motor, dual limit switches, and a median-filtered hall sensor to measure and compensate about 250 steps of mechanical backlash
- Validated positioning across 84 test runs measured with a dial indicator accurate to 0.0002 in, with all 40 trials of 1 mil moves landing within ±0.2 mil, 2.5x tighter than the 0.5 mil increment spec, and mean error under 0.03 mil in both directions

UC Davis Undergraduate Level Research Program July 2024 – Aug 2024

Researcher – Physics & Engineering Cluster *Davis, CA*

- Applied nuclear and electrical engineering principles to analyze real-world systems, performing quantitative evaluations of fission and fusion processes, critical mass requirements, reactor design constraints, and electro-optical device behavior
- Executed laboratory investigations using interferometry, laser systems, optical waveguides, and fiber-optic setups to characterize electromagnetic propagation, assess signal transmission behavior, and interpret experimental results

Projects

Two-Way Pager | *Hands-On PCB Engineering (EE 198)* Fall 2026

- Designing a 2-layer KiCad PCB for battery-powered pagers that text over the 915 MHz ISM band, pairing an ESP32-S3 with a CC1101 transceiver over SPI to target a 2 km campus range under a $100 budget for two boards

Leadership

IEEE at Berkeley May 2026 – Present

Co-Director, Industry Relations *Berkeley, CA*

- Grew Industry Relations from a 5-person subteam into a 10+ member committee, creating LinkedIn, restaurant, alumni mixer, and advertising subteams and redoing the sponsorship packet to increase IEEE revenue and company partnerships
- Connect students with recruiters from 10+ partner companies including Apple, Intel, TSMC, and Roblox by coordinating info sessions, sponsorships, and company field trips and facilitating direct networking with industry professionals at each event
- Manage 8 subcommittees spanning company outreach, correspondence, event logistics, advertising, poster making, LinkedIn, alumni mixers, and restaurant partnerships, running recruiting events end to end from room booking to day-of hosting